

Philosophy Day

Mathematical Logic | 23 October 2025

ACTIVITY: Philosophical questions.

What are your thoughts on the following questions?

1. We saw that all mathematical objects– real numbers, geometric objects, functions– can be thought of as complicated sets. Is this actually “true”? Or are these just to be thought of as encodings of the objects, which already exist?
2. Do our considerations about infinity have any bearing on the real world? For example, is the set of objects in the universe countable, uncountable, or does it not make sense to ask?
3. The Banach-Tarski paradox shows that a sphere can be disassembled into 5 pieces which can be re-assembled into two copies of the same sphere. These pieces are infinitely detailed, so cannot be physical. Does this diminish the paradox at all?
4. There are some statements, like the continuum hypothesis, which cannot be proven or disproven from the commonly accepted axioms of mathematics. Does it make sense to ask whether these statements are “true” or “false”?